

House Price Prediction using Regression

A Machine Learning Approach

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Subject - Introduction To AI(AI101B)





Objective

Price Prediction

Estimate home sale prices with ML

Feature Analysis

Identify key price influencers

Model Comparison

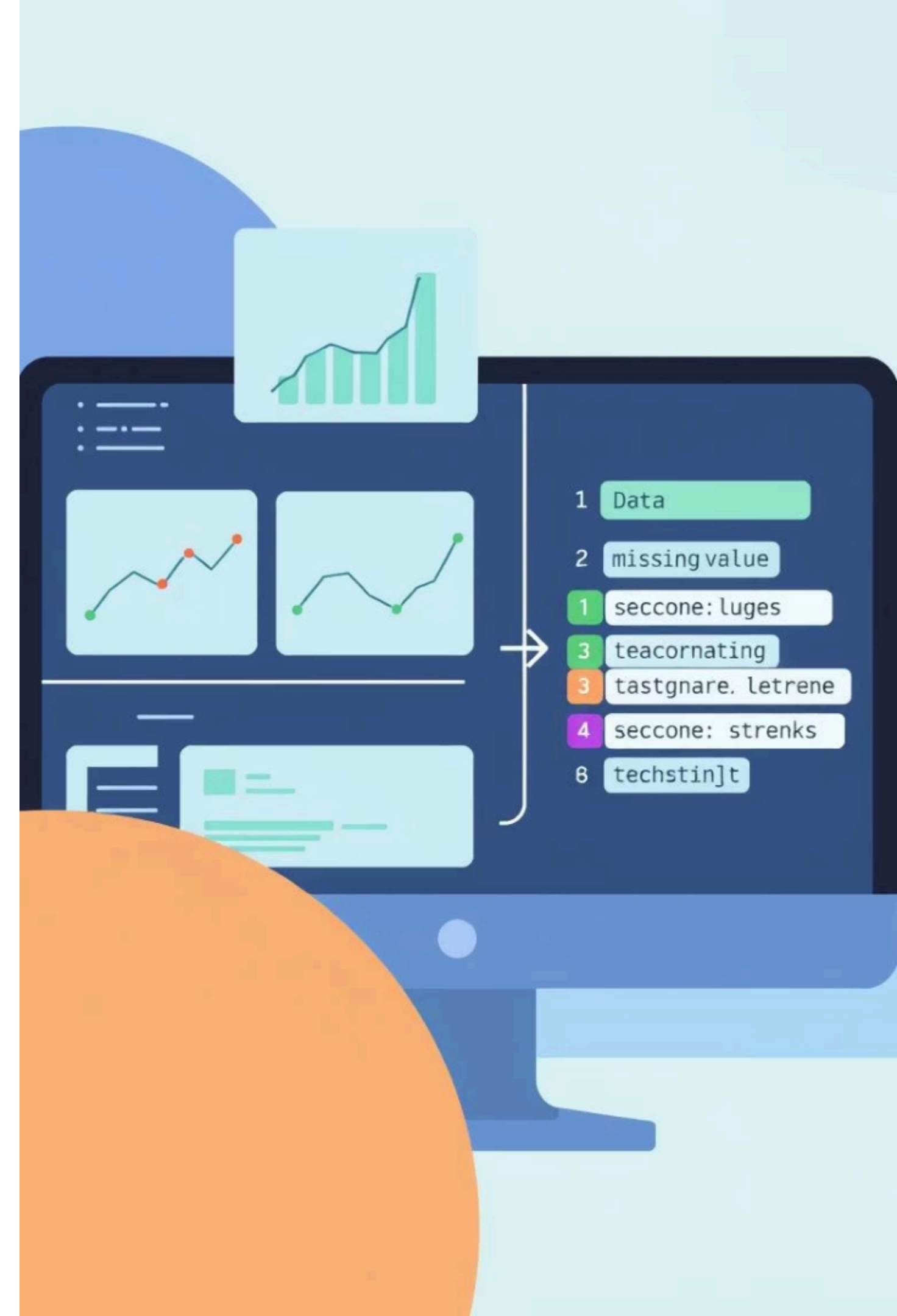
Evaluate regression model performance

Dataset Overview

Source	Size	Target Variable	Features
Kaggle - Ames Housing Dataset	1,460 rows, 81 columns	SalePrice	Area, Quality, Year, and more

Data Preprocessing

- 1 Missing Data**
Imputed using median and mode
- 2 Encoding**
Label encoded categorical variables
- 3 Scaling**
Feature scaling for regression input
- 4 Data Split**
80% training, 20% testing



Algorithms Used



Linear Regression

Baseline model



Ridge Regression

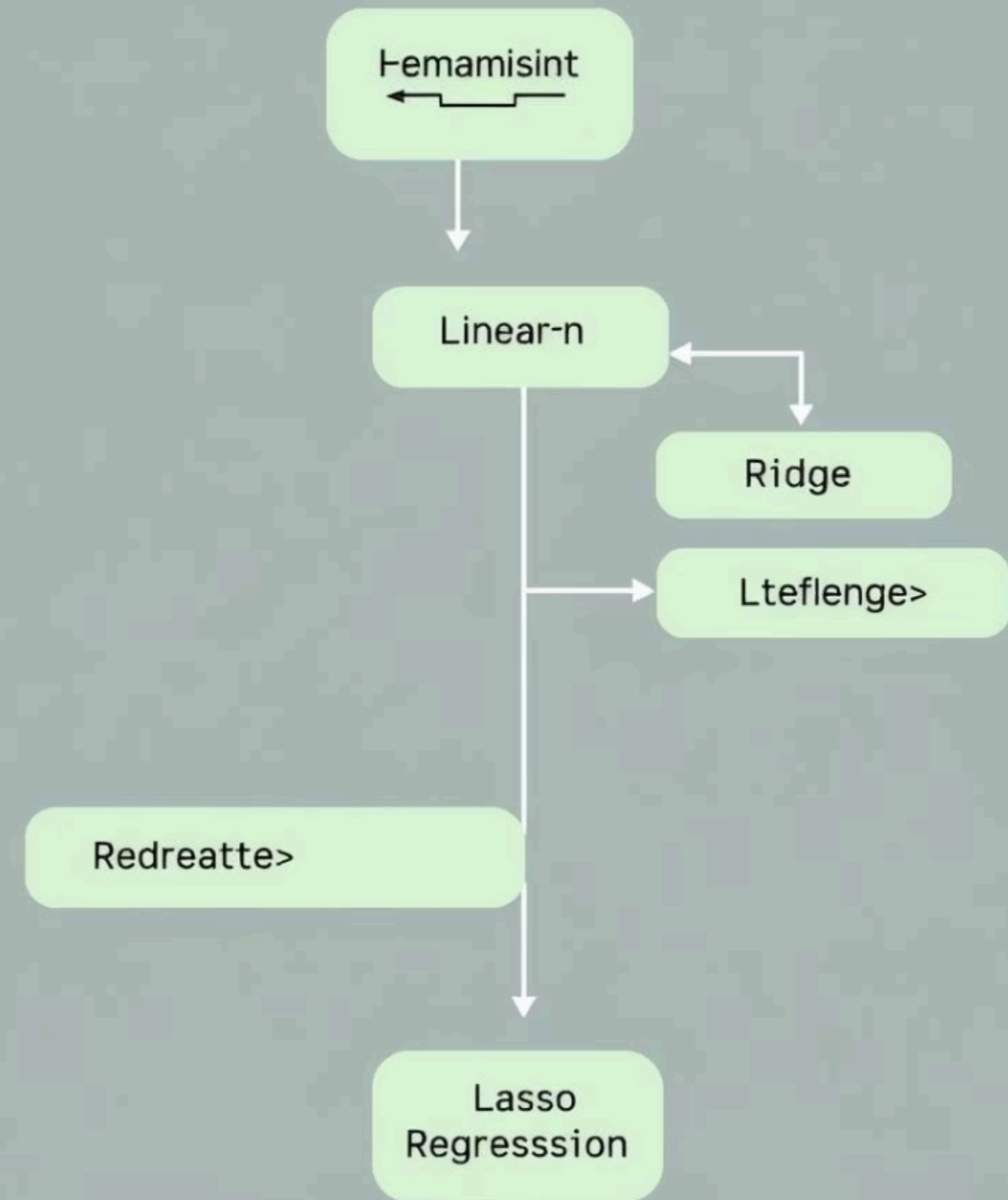
L2 regularization



Lasso Regression

L1 regularization and feature selection

Implemented with scikit-learn library



Evaluation Metrics

MAE

Mean Absolute Error

MSE

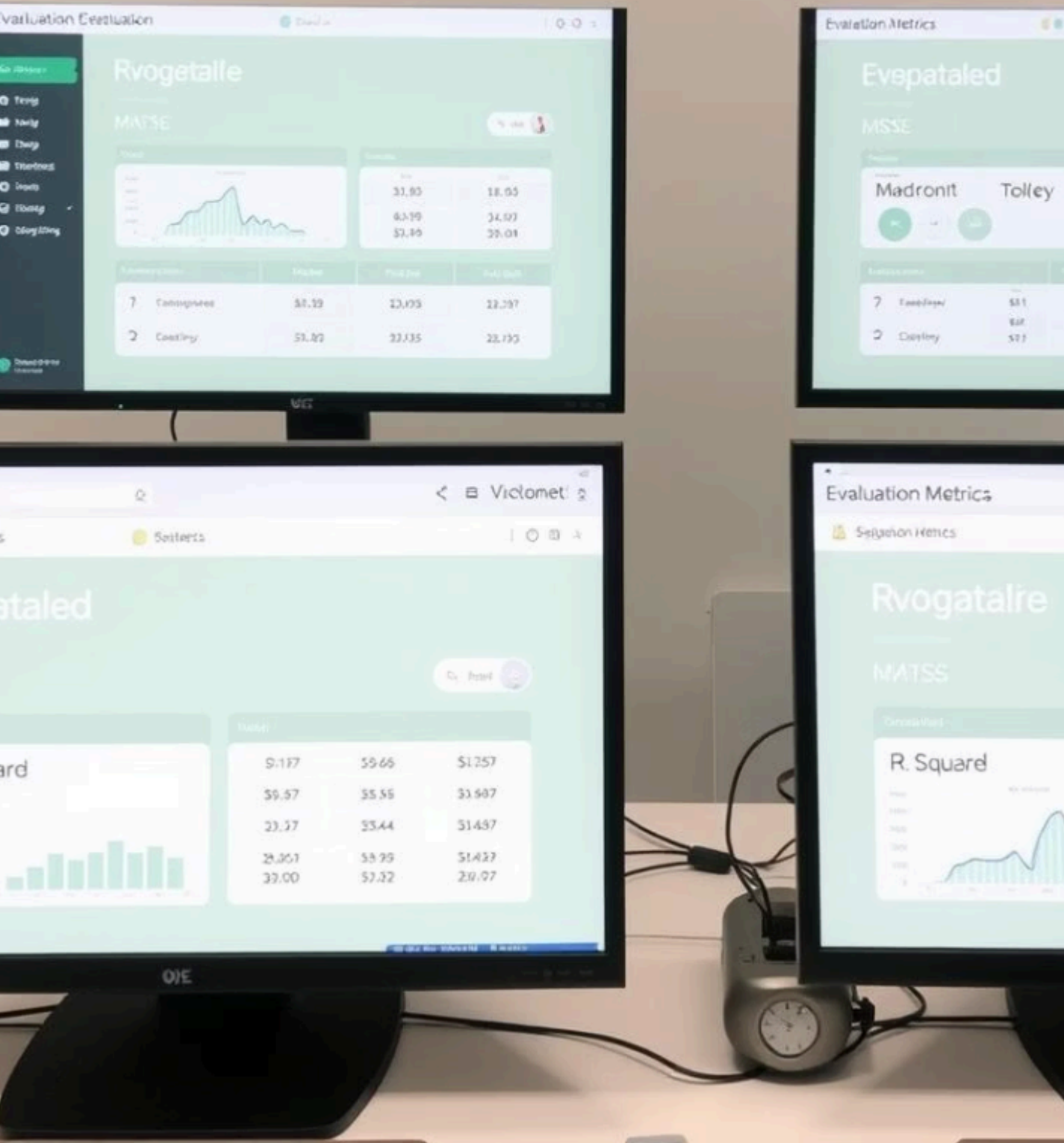
Mean Squared Error

RMSE

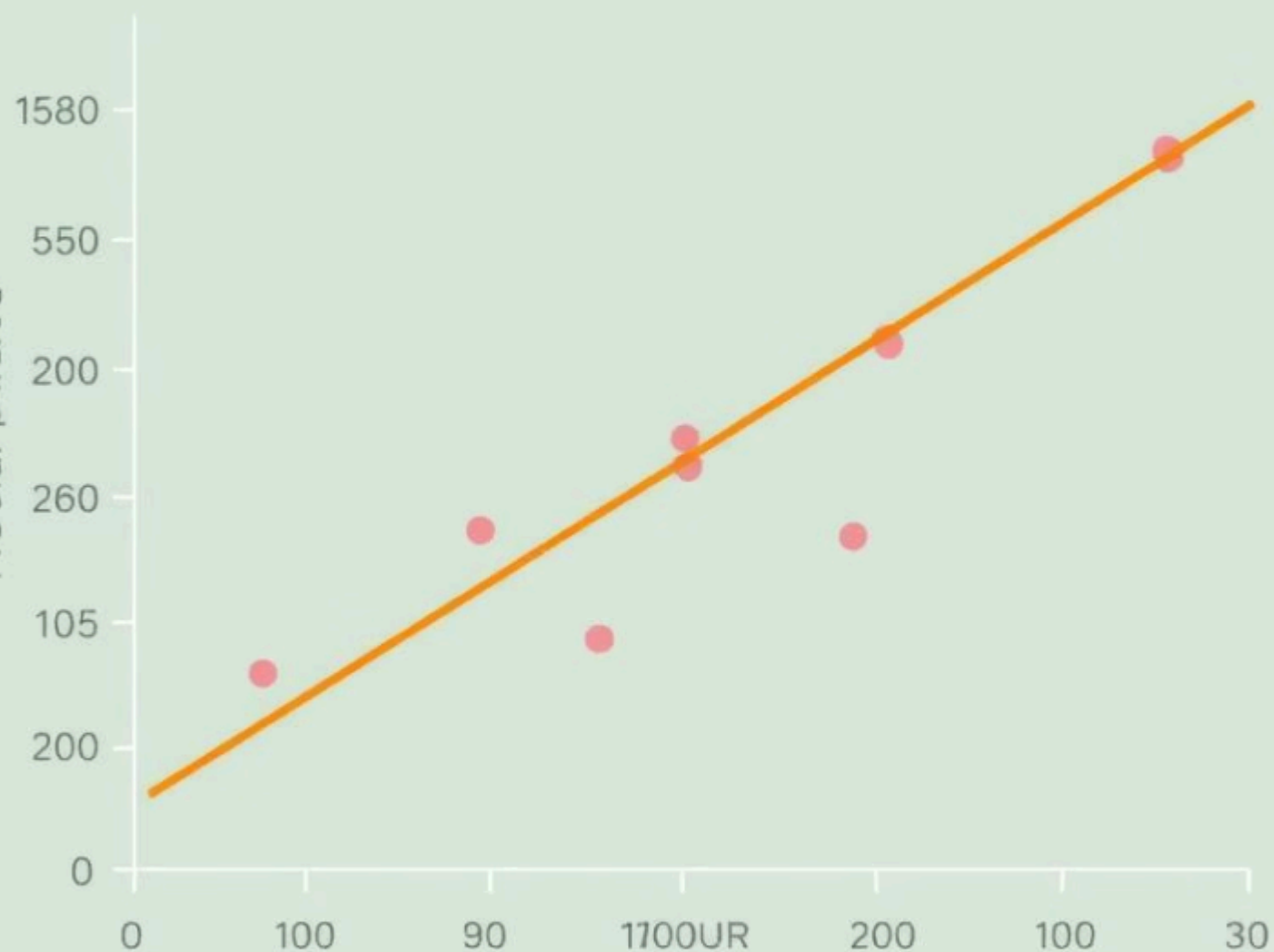
Root Mean Squared Error

R^2 Score

Goodness of fit measure



Houness precees house prices



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Results - Linear Regression



Performance

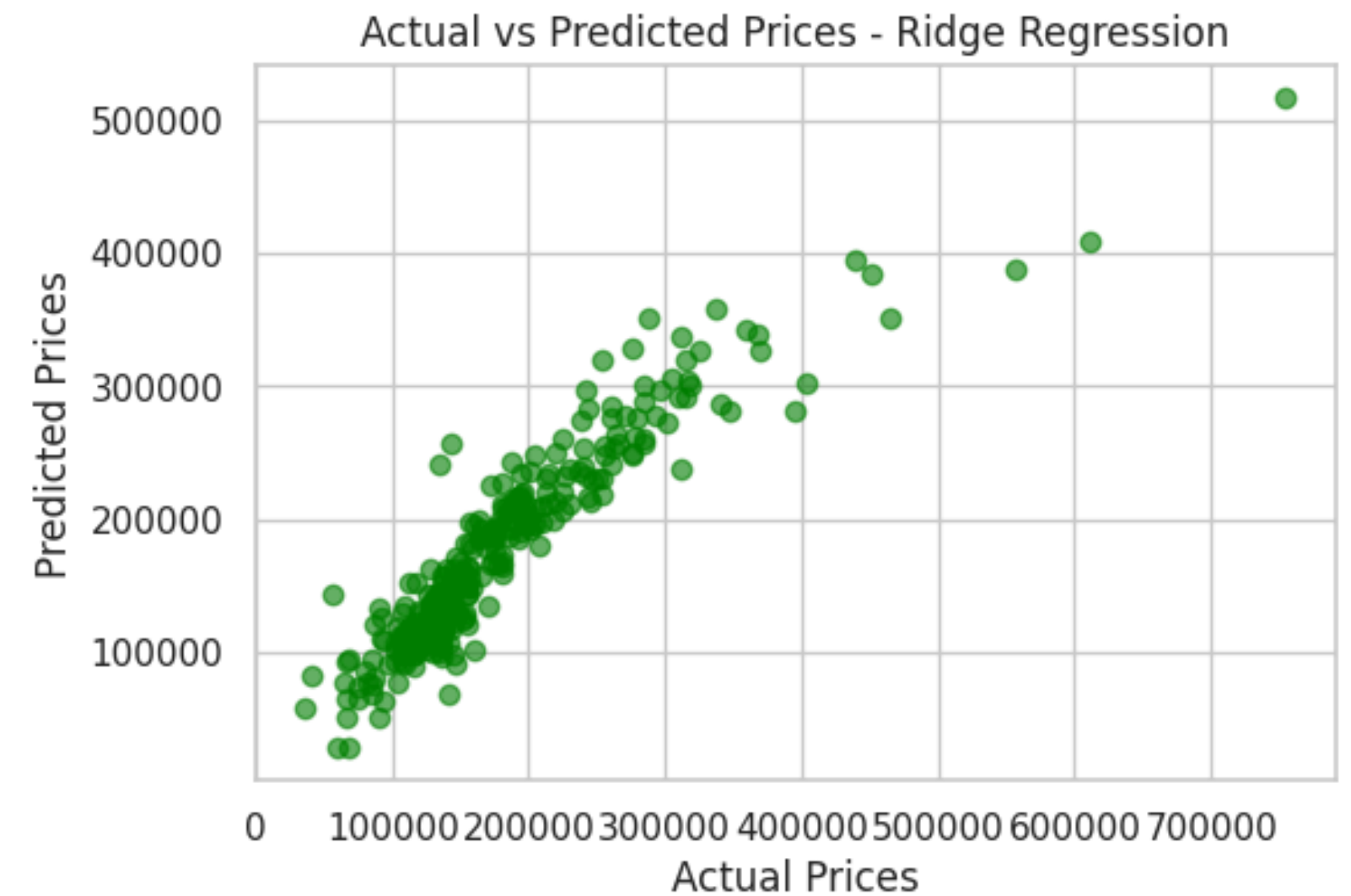
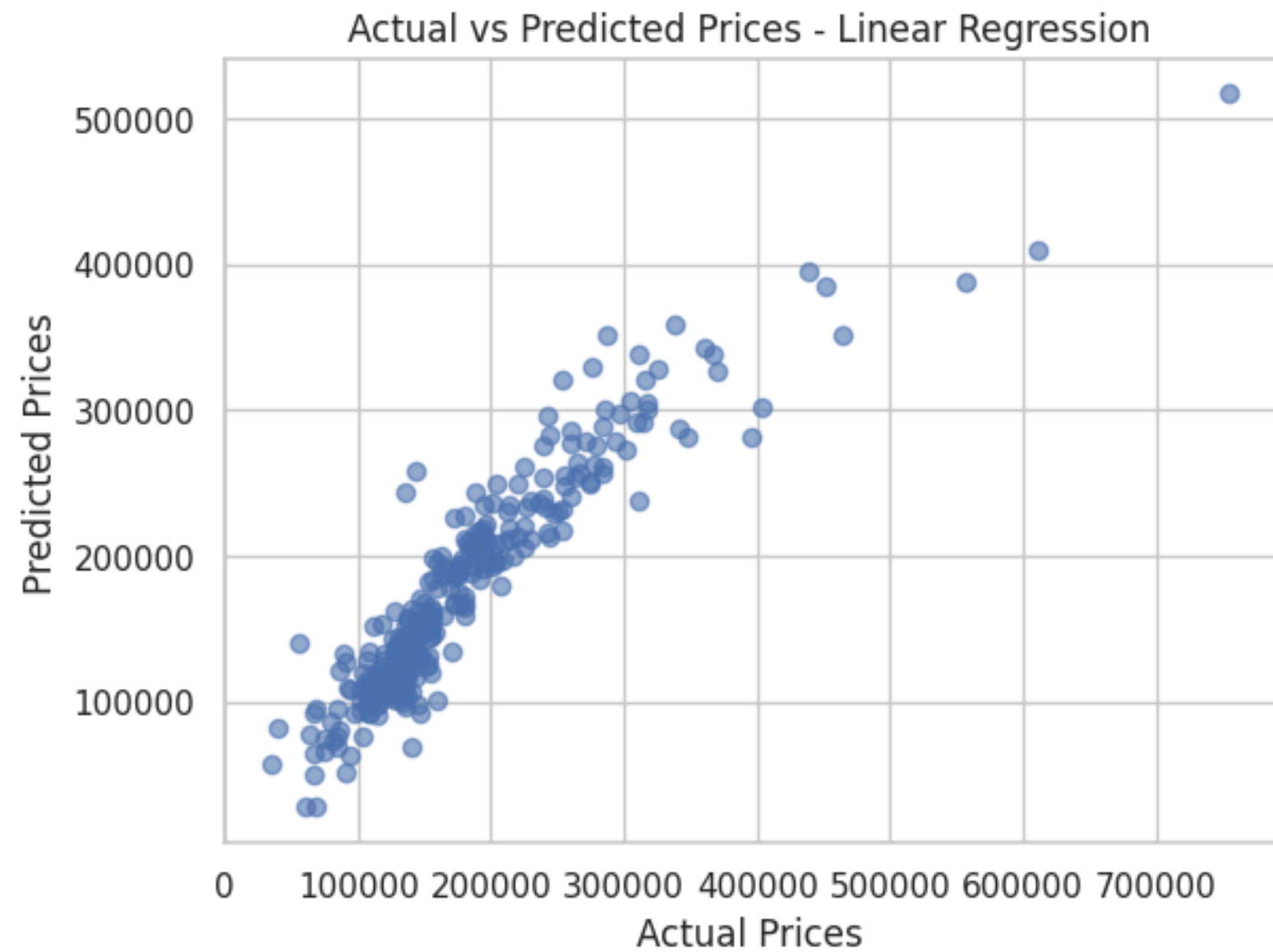
R^2 : [Insert value]



Error

RMSE: [Insert value]

Results - Ridge & Lasso Regression



Results



Enter the following details:

Overall Quality (1-10): 5

Above Grade Living Area (sq ft): 45

Garage Cars: 2

Total Basement Area (sq ft): 80

Number of Full Bathrooms: 4

Year Built: 2020



Predicted House Price: \$56,732.94



Conclusion

- Ridge regression achieved best accuracy
- Lasso enhances model interpretability
- Regression reliable for price prediction



Thank You

Questions and Discussion